reflex was first demonstrated by the Russian physiologist **Pavlov** in his classical conditioning experiment in a dog. The cerebral cortex controls the conditioned reflex.

**Peripheral Neural System (PNS)**

PNS consists of all nervous tissue outside the CNS. Components of PNS include nerves, ganglia, enteric plexuses and sensory receptors. A nerve is a chord like structure that encloses several neurons inside. Ganglia (singular-ganglion) are small masses of nervous tissue, consisting primarily of neuron cell bodies and are located outside the brain and spinal cord. Enteric plexuses are extensive networks of neurons located in the walls of organs of the gastrointestinal tract. The neurons of these plexuses help in regulating the digestive system. The specialized structure that helps to respond to changes in the environment i.e. stimuli are called **sensory receptor** which triggers nerve impulses along the afferent fibres to CNS. PNS comprises of cranial nerves arising from the brain and spinal nerves arising from the spinal cord.

**(A) Cranial Nerves:** There are 12 pairs of cranial nerves, of which the first two pairs arise from the fore brain and the remaining 10 pairs from the mid brain. Other than the Vagus nerve, which extends into the abdomen, all cranial nerves serve the head and face.

**(B) Spinal Nerves:** 31 pairs of spinal nerves emerge out from the spinal cord through spaces called the **intervertebral foramina** found between the adjacent vertebrae. The spinal nerves are named according to the region of vertebral column from which they originate

i. Cervical nerves (8 pairs)
ii. Thoracic nerves (12 pairs)
iii. Lumbar nerves (5 pairs)
iv. Sacral nerves (5 pairs)
v. Coccygeal nerves (1 pair)

Each spinal nerve is a mixed nerve containing both afferent (sensory) and efferent (motor) fibres**.** It originates as two roots:1) a posterior dorsal root with a ganglion outside the spinal cord and 2) an anterior ventral root with no external ganglion.

**Somatic Neural System** (SNS)

The **somatic neural system** (SNS or voluntary neural system) is the part of the peripheral neural system associated with the voluntary control of body movements via skeletal muscles. The somatic neural system comprises of the sensory and motor nerves. Major functions of the somatic neural system include voluntary movement of the muscles and organs, and reflex movements.

In adult, the total CSF volume is about 150 ml and is replaced every 8 hours. About 500 ml of CSF is formed daily. The choroid plexus helps cleanse the CSF by removing waste products.

**Autonomic Neural System**

The autonomic neural system is auto functioning and self governed. It is a part of peripheral neural system that innervates smooth muscles, glands and cardiac muscle. This system controls and coordinates the involuntary activities of various organs. ANS controlling centre is in the hypothalamus.

Autonomic neural system comprises the following components:

**Preganglionic neuron** whose cell body is in the brain or spinal cord; its myelinated axon exits the CNS as part



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